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# Operating Systems
## Case Study

Q1] 3 page frames

1, 2, 1, 3, 7, 4, 5, 6, 3, 1, 7, 4, 5, 6, 1, 2, 1, 3, 7, 4, 5, 4, 5, 3, 1, 4, 5, 6, 7, 4, 3

1) LRU

| | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 7 | 7 | 7 | 6 | 6 | 6 | 7 | 7 | 7 | 6 | 6 | 6 | 6 | 3 | 3 | 3 | 5 | 5 | 5 | 5 | 5 | 4 | 4 | |
| | 2 | 2 | 2 | 2 | 4 | 4 | 4 | 3 | 3 | 3 | 4 | 4 | 4 | 1 | 1 | 1 | 1 | 7 | 7 | 7 | 7 | 7 | 3 | 3 | 3 | 5 | |
| | | 3 | 3 | 3 | 5 | 5 | 5 | 1 | 1 | 1 | 5 | 5 | 5 | 2 | 2 | 2 | 2 | 4 | 4 | 4 | 4 | 4 | 1 | 1 | 1 | | |

| | | | |
|---|---|---|---|
| 4 | 7 | 7 | 7 |
| 5 | 5 | 4 | 4 |
| 6 | 6 | 6 | 3 |

no. of page faults = 27

2) Optimal

| | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 4 |
| | 2 | 2 | 2 | 7 | 4 | 5 | 6 | 6 | 6 | 7 | 4 | 5 | 6 | 6 | 2 | 2 | 2 | 7 | 4 | 5 | 4 | 5 | 5 | 5 | 5 | | |
| | | | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | | |

| | | | | |
|---|---|---|---|---|
| 4 | 4 | 4 | 4 | 4 |
| 5 | 6 | 7 | 7 | 7 |
| 3 | 3 | 3 | 3 | 3 |

no. of page faults = 18

3) FIFO

| | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 7 | 7 | 7 | 6 | 6 | 6 | 7 | 7 | 7 | 6 | 6 | 6 | 6 | 3 | 3 | 3 | 5 | 5 | 5 | 5 | 5 | 4 | | |
| | 2 | 2 | 2 | 2 | 4 | 4 | 4 | 3 | 3 | 3 | 4 | 5 | 4 | 4 | 1 | 1 | 1 | 1 | 7 | 7 | 7 | 7 | 3 | 3 | 3 | | |
| | | | 3 | 3 | 3 | 5 | 5 | 5 | 1 | 1 | 1 | 5 | 5 | 5 | 2 | 2 | 2 | 2 | 4 | 4 | 4 | 4 | 4 | 1 | 1 | | |

| | | | | |
|---|---|---|---|---|
| 4 | 4 | 7 | 7 | 7 |
| 5 | 5 | 5 | 4 | 4 |
| 1 | 6 | 6 | 6 | 3 |

no. of page faults = 27

4) LFU counter.

| | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| | 2 | 2 | 2 | 7 | 7 | 5 | 5 | 3 | 3 | 3 | 4 | 4 | 6 | 6 | 6 | 6 | 6 | 3 | 3 | 4 | 4 | 4 | 4 | 4 | 4 |
| | | | 3 | 3 | 4 | 4 | 6 | 6 | 6 | 7 | 7 | 5 | 5 | 5 | 2 | 2 | 2 | 7 | 7 | 5 | 5 | 5 | 5 | 5 | 5 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 3 | 3 | 3 | 5 | 5 | 5 | 5 | 3 |
| 5 | 5 | 4 | 4 | 6 | 7 | 4 | 4 |

no. of page faults = 24

5) MFU

| | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 7 | 4 | 4 | 6 | 6 | 6 | 6 | 4 | 5 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 |
| | 2 | 2 | 2 | 2 | 2 | 5 | 5 | 5 | 1 | 7 | 7 | 7 | 7 | 1 | 2 | 1 | 1 | 7 | 7 | 5 | 5 | 5 | 5 | 5 | 5 |
| | | | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 4 | 4 | 4 | 4 | 4 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 |
| 3 | 1 | 1 | 5 | 5 | 7 | 7 | 3 |
| 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |

no. of page faults = 276

6) second chance page replacement algorithm

| | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 5 | 5 | 5 | 1 | 1 | 1 | 5 | 5 | 5 | 2 | 2 | 2 | 7 | 7 | 7 | 7 | 7 | 3 | 1 | 1 | 1 |
| | 2 | 2 | 2 | 7 | 7 | 7 | 6 | 6 | 6 | 7 | 7 | 7 | 6 | 6 | 6 | 6 | 3 | 3 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| | | | 3 | 3 | 4 | 4 | 4 | 3 | 3 | 3 | 4 | 4 | 4 | 1 | 1 | 1 | 1 | 1 | 1 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |

| | | | |
|---|---|---|---|
| 6 | 6 | 6 | 3 |
| 4 | 7 | 7 | 7 |
| 5 | 5 | 4 | 4 |

no. of page faults = 25

Q2] virtual address size = 46 bits
page table entry size = 32 bits = 4 bytes
no. of translation levels : 3 (T1, T2, T3)
let page size = $2^P$ bytes
P = no. of bits required for the page offset

$$\text{entries per page} = \frac{\text{page size}}{\text{PTE size}} = \frac{2^P}{2^2} = 2^{P-2}$$

bits for T1 index = p-2
bits for T2 index = p-2
bits for T3 index = p-2
bits for page offset = p

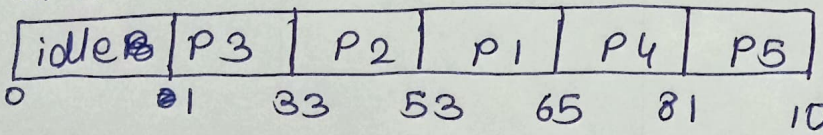
$$3 \times (p-2) + p = 46$$
$$4p - 6 = 46$$
$$4p = 52$$
$$p = 13$$

$\therefore$ page size $= 2^p = 2^{13}$ bytes
$= \underline{8\ KB}$

Q3]

| process | arrival time (ms) | total execution time (ms) | CPU burst | IIO burst |
|---|---|---|---|---|
| P1 | 3 | 15 | 12 | 3 |
| P2 | 2 | 25 | 20 | 5 |
| P3 | 1 | 40 | 32 | 8 |
| P4 | 5 | 20 | 16 | 4 |
| P5 | 10 | 30 | 24 | 6 |

1) FCFS




~~waiting time~~

turn around time

P1 : 65+3-3 = 65 ms
P2 : 53+5-2 = 56 ms
P3 : 33+8-1 = 40 ms
P4 : 81+4-5 = 80 ms
P5 : 105+6-10 = 101 ms

waiting time

P1 : 65-12-3 = 50 ms
P2 : 53-20-2 = 31 ms
P3 : 33-32-1 = 0 ms
P4 : 81-16-5 = 60 ms
P5 : 105-25-10 = 71 ms

average WT = $\frac{50+31+0+60+71}{5}$ = 42.4 ms

average TAT = $\frac{65+56+40+80+101}{5}$ = 68.4 ms

2) SJF Preemptive

| idle | P3 | P2 | P4 | P4 | P1 | P2 | P5 | P3 |
0 1 2 5 10 21 33 50 74 105

Turn around time

P1: 33 - 3 = 30 ms
P2: 50 - 2 = 48 ms
P3: 105 - 1 = 104 ms
P4: 21 - 5 = 16 ms
P5: 74 - 10 = 64 ms

waiting time

P1: 30 - 12 = 18 ms
P2: 48 - 20 = 28 ms
P3: 104 - 32 = 72 ms
P4: 16 - 16 = 0 ms
P5: 64 - 24 = 40 ms

average TAT = $\frac{30+48+104+16+64}{5}$ = 52.4 ms

average WT = $\frac{18+28+72+0+40}{5}$ = 31.6 ms

3) Round Robin (time slice = 3ms)

| idle | P3 | P2 | P1 | P3 | P4 | P2 | P5 | P1 | P3 | P4 | P2 | P5 | P1 | P3 |
0 1 4 7 10 13 16 19 22 25 28 31 34 37 40 43

| P4 | P2 | P5 | P1 | P3 | P4 | P2 | P5 | P3 | P4 | P2 | P5 | P3 |
43 46 49 52 55 58 61 64 67 70 73 76 79 82

| P4 | P2 | P5 | P3 | P5 | P3 | P5 | P3 |
82 83 85 88 91 94 97 100 105

| | TAT (ms) | WT (ms) |
|---|---|---|
| P1 | 55 | 40 |
| P2 | 88 | 63 |
| P3 | 112 | 72 |
| P4 | 82 | 62 |
| P5 | 96 | 66 |

avg TAT = 86.6 ms
avg WT = 60.6 ms

Q4] main memory access time = 110 ns
TLB lookup time = 3 ns
TLB hit ratio = 94% = 0.94
TLB miss ratio = 6% = 0.06
page fault rate = 0.2% = 0.002
page fault service time = 7 ms
no. of page table levels = 3

i) Effective access time without TLB

$$T_{normal} = (3+1) \times 110$$
$$= 440 \text{ ns}$$

$$EAT = (1-p) \times T_{normal} + p \times T_{pagefault}$$
$$= (1-0.002) \times 440 + 0.002 \times 7 \times 10^6$$
$$= 14439.12 \text{ ns}$$

ii) Effective access time with TLB

case 1: TLB hit

$$T_{hit} = 3 + 110 = 113 \text{ ns}$$

case 2: TLB miss

$$T_{miss} = 3 + (3 \times 110) + 110$$
$$= 443 \text{ ns}$$

$$T_{avg} = h T_{hit} + (1-h) T_{miss}$$
$$= 0.94 \times 113 + 0.06 \times 443$$
$$= 132.8 \text{ ns}$$

$$EAT = (1-p) T_{avg} + p T_{pagefault}$$
$$= 0.998 \times 132.8 + 0.002 \times 7 \times 10^6$$
$$= 14132.5344 \text{ ns}$$

iii) Percentage improvement in EAT

$$\text{improvement} = \frac{EAT_{\text{without TLB}} - EAT_{\text{with TLB}}}{EAT_{\text{without TLB}}} \times 100$$

$$= \frac{14439.12 - 14132.5344}{14439.12} \times 100$$

$$= 2.12\%.$$

iv) High TLB hit ratio means that most virtual to physical address translations are found directly in TLB, avoiding the need to access the page table in main memory.

On a TLB hit: only the TLB lookup (3 ns) and the actual memory access (110 ns) are required, giving 113 ns

On a TLB miss: three page table accesses and one data access are required, in addition to the TLB lookup, giving 443 ns

with a 94% hit ratio: most memory references take the faster path, reducing the average access time from 440 ns to 132.8 ns before accounting for page faults.

The overall improvement is only about 2.12% as in this particular problem because the page fault service time is extremely large (7 ms). Even a page fault rate of just 0.2% adds approximately 14000 ns to the expected access time, dominating the normal memory access cost.

Q5] · Q1 = Round robin TQ = 2 ms
Q2 = Round robin TQ = 3 ms
Q3 = FCFS
Q4 = shortest job first (SJF)
Q5 = Round robin TQ = 4 ms
Priority: Q1 > Q2 > Q3 > Q4 > Q5

| Process | AT (ms) | BT (ms) | Queue |
|---|---|---|---|
| P1 | 0 | ~~7~~ 0 | Q3 |
| P2 | 1 | ~~4~~ ~~2~~ 0 | Q1 |
| P3 | 2 | 6 0 | Q4 |
| P4 | 3 | ~~5~~ ~~2~~ 0 | Q2 |
| P5 | 4 | ~~3~~ ~~1~~ 0 | Q1 |
| P6 | 5 | ~~8~~ ~~4~~ 0 | Q5 |
| P7 | 6 | 4 0 | Q3 |
| P8 | 7 | ~~6~~ ~~3~~ 0 | Q2 |
| P9 | 8 | ~~5~~ 0 | Q4 |
| P10 | 9 | ~~3~~ 0 | Q5 |

a)
Ready queue

Q1 | ~~P2~~ (1) | ~~P5~~ (4) | ~~P2~~ | ~~P5~~
Q2 | ~~P4~~ (3) | ~~P8~~ (7) | ~~P4~~ | ~~P8~~
Q3 | ~~P1~~ (0) | ~~P7~~ (6)
Q4 | ~~P3~~ (2) | ~~P9~~ (8)
Q5 | ~~P6~~ (5) | ~~P10~~ (9) | P6

b)
Gantt chart

| P1 | P2 | P5 | P2 | P5 | P4 | P8 | P4 | P8 | P7 |
0 7 9 11 13 14 17 20 22 25 29

| P9 | P3 | P6 | P10 | P6 |
29 34 40 44 47 51

b)

~~Process~~ ~~CT~~ ~~TAT~~ ~~WT~~

Gantt Chart:

| P1 | P2 | P5 | P4 | P8 | P4 | P8 | P1 | P7 | P9 |
0 1 5 8 11 14 16 19 25 29 34

| P3 | P6 | P10 | P6 |
34 40 44 47 51

c)

| | CT(ms) | TAT (ms) | WT (ms) |
|---|---|---|---|
| P1 | 25 | 25-0 = 25 | 25-7 = 18 |
| P2 | 5 | 5-1 = 4 | 4-4 = 0 |
| P3 | 40 | 40-2 = 38 | 38-6 = 32 |
| P4 | 16 | 16-3 = 13 | 13-5 = 8 |
| P5 | 8 | 8-4 = 4 | 4-3 = 1 |
| P6 | 51 | 51-5 = 46 | 46-8 = 38 |
| P7 | 29 | 29-6 = 23 | 23-4 = 19 |
| P8 | 19 | 19-7 = 12 | 12-6 = 6 |
| P9 | 34 | 34-8 = 26 | 26-5 = 21 |
| P10 | 47 | 47-9 = 38 | 38-3 = 35 |

TAT = CT - AT

WT = TAT - BT

d) average TAT = $\frac{229}{10}$ = 22.9 ms

average WT = $\frac{178}{10}$ = 17.8 ms

Q6] Hole size (KB) = 6, 22, 14, 30, 9, 17, 11, 26
Process size (KB) = 12, 7, 20, 10, 15, 24

total initial free memory = 135 KB
total memory requested by all processes = 88 KB

1) First Fit

| | size | selected hole | hole size (KB) before | hole size (KB) ~~aft~~ remaining |
|---|---|---|---|---|
| P1 | 12 KB | H2 | 22 | 10 |
| P2 | 7 KB | H2 | 10 | 3 |
| P3 | 20 KB | H4 | 30 | 10 |
| P4 | 10 KB | H3 | 14 | 4 |
| P5 | 15 KB | H6 | 17 | 2 |
| P6 | 24 KB | H8 | 26 | 2 |

2) Best Fit

| | size (KB) | selected hole | hole size (KB) before | hole size (KB) remaining |
|---|---|---|---|---|
| P1 | 12 ~~KB~~ | H3 | 14 | 2 |
| P2 | 7 | H5 | 9 | 2 |
| P3 | 20 | H2 | 22 | 2 |
| P4 | 10 | H7 | 11 | 1 |
| P5 | 15 | H6 | 17 | 2 |
| P6 | 24 | H8 | 26 | 2 |

3) Worst Fit

| | size (KB) | selected hole | hole size (KB) before | hole size (KB) remaining |
|---|---|---|---|---|
| P1 | 12 | H4 | 30 | 18 |
| P2 | 7 | H8 | 26 | 19 |
| P3 | 20 | H2 | 22 | 2 |
| P4 | 10 | H8 | 19 | 9 |
| P5 | 15 | H4 | 18 | 3 |
| P6 | 24 | cant select | — | — |

| | First fit | Best fit | Worst fit |
|---|---|---|---|
| total initialized memory | 135 KB | 135 KB | 135 KB |
| total allocated memory | 88 KB | 88 KB | 64 KB |
| internal fragmentation | 0 KB | 0 KB | 0 KB |
| total free memory | 47 KB | 47 KB | 71 KB |
| largest remaining hole | 11 KB | 30 KB | 17 KB |
| external fragmentation | 47 KB | 47 KB | 71 KB |
| process not allocated | — | — | P6 (24KB) |

Calculations:

first fit & best fit

allocated = 12+7+20+10+15+24 = 88 KB

free = 135 - 88 = 47 KB

worst fit

allocated = 12+7+20+10+15 = 64 KB

free = 135 - 64 = 71 KB

$$\text{memory utilization} = \frac{\text{total allocated memory}}{\text{total available memory}} \times 100$$

$$\text{first fit} = \frac{88}{135} \times 100 = 65.19\%.$$

$$\text{best fit} = \frac{88}{135} \times 100 = 65.19\%.$$

$$\text{worst fit} = \frac{64}{135} \times 100 = 47.41\%.$$

Q7] no. of pages = 16

page size = 1 KB = $2^{10}$ bytes

| page no. | frame no |
|---|---|
| 0 | 5 |
| 1 | 9 |
| 2 | 3 |
| 3 | 7 |
| 4 | 6 |
| 5 | 1 |

logical

a) ~~physical~~ address = 2504

page no. (p) = $\left\lfloor \frac{2504}{1024} \right\rfloor = 2$

offset (d) = 2504 % 1024 = 456

frame no. (f) = 3

physical address = (3 × 1024) + 456 = 3528

b) logical address = 4113

page no = $\left\lfloor \frac{4113}{1024} \right\rfloor = 4$

offset = 4113 % 1024 = 17

frame no. = 6

physical address = (6 × 1024) + 17 = 6161

c) logical address = 1124

page no. = $\left\lfloor \frac{1124}{1024} \right\rfloor = 1$

offset = 1124 % 1024 = 100

frame no. = 9

physical address = (9 × 1024) + 100 = 9316

d) logical address = 2051

page no. = $\left\lfloor \frac{2051}{1024} \right\rfloor = 2$

offset = 2051 % 1024 = 3

frame no. = 3

physical address = (3 × 1024) + 3 = 3075

e) logical address = 3124

page no. = $\left\lfloor \frac{3124}{1024} \right\rfloor = 3$

offset = 3124 % 1024 = 52

frame no. = 7

physical address = (7 × 1024) + 52 = 7220

f) logical address = 5001

page no. = $\left\lfloor \frac{5001}{1024} \right\rfloor = 4$

offset = 5001 % 1024 = 905

frame no. = 6

physical address = (6 × 1024) + 905 = 7049

Q8] page size = 32KB = $2^{15}$ bytes

page offset = 15 bits

virtual memory address size = 48 bits

physical memory size = 2TB = $2^{41}$ bytes

physical address = 41 bits

page table entry (PTE) size = 8 bytes = $2^3$ bytes

1) no. of levels of page tables required

page no. bits = 48 − 15 = 33 bits

entries per page table = $\frac{\text{page size}}{\text{PTE size}} = \frac{2^{15}}{2^3} = 2^{12}$ entries

no. of levels = $\left\lceil \frac{33}{12} \right\rceil$ = 3 levels

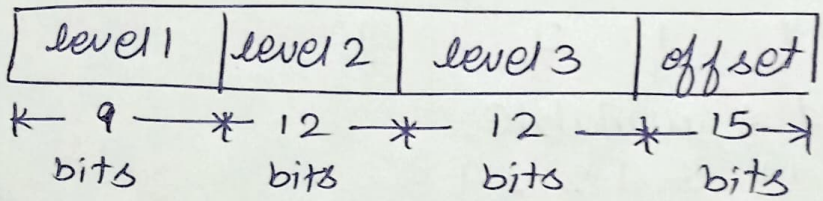
2) division of virtual address

level 1 = 33 − 12 − 12 = 9 bits

level 2 = 12 bits

level 3 = 12 bits

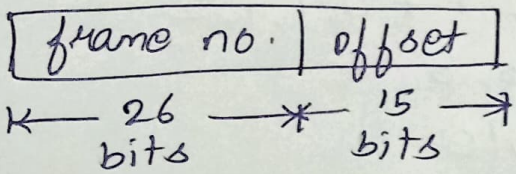
page offset = 15 bits




3) no. of bits required for the frame ~~no.~~ no.

= 41 − 15 = 26 bits

4) division of physical address




5) total no. of page frames available

= $\frac{\text{physical memory size}}{\text{page size}} = \frac{2^{41}}{2^{15}} = 2^{26}$

Q9]

| | allocation | | | | max | | | | available | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | A | B | C | D | A | B | C | D | A | B | C | D |
| T0 | 0 | 1 | 1 | 2 | 0 | 2 | 2 | 3 | 1 | 2 | 2 | 1 |
| T1 | 1 | 0 | 2 | 0 | 2 | 2 | 4 | 2 | | | | |
| T2 | 1 | 2 | 3 | 1 | 3 | 4 | 4 | 3 | | | | |
| T3 | 1 | 0 | 1 | 2 | 2 | 2 | 2 | 3 | | | | |
| T4 | 0 | 1 | 2 | 1 | 1 | 3 | 3 | 2 | | | | |

a) need matrix

need = max - allocation

| | A | B | C | D |
|---|---|---|---|---|
| T0 | 0 | 1 | 1 | 1 |
| T1 | 1 | 2 | 2 | 2 |
| T2 | 2 | 2 | 1 | 2 |
| T3 | 1 | 2 | 1 | 1 |
| T4 | 1 | 2 | 1 | 1 |

b) T0: need ≤ available

0 1 1 1 ≤ 1 2 2 1

↳ true

∴ T0 can be executed

available = available + allocation

= 1, 2, 2, 1 + 0, 1, 1, 2

= 1, 3, 3, 3

T1: need ≤ available

1, 2, 2, 2 ≤ 1, 3, 3, 3

↳ true

∴ T1 can be executed

available = available + allocation

= 1, 3, 3, 3 + 1, 0, 2, 0

= 2, 3, 5, 3

T2: need ≤ available

2, 2, 1, 2 ≤ 2, 3, 5, 3

↳ true

∴ T2 can be executed

available = available + ~~executed~~ allocation

= 2, 3, 5, 3 + 1, 2, 3, 1

= 3, 5, 8, 4

T3: need ≤ available

1, 2, 1, 1 ≤ 3, 5, 8, 4

↳ true

∴ & T3 can be executed

available = available + allocation

= 3, 5, 8, 4 + 1, 0, 1, 2

= 4, 5, 9, 6

T4: need ≤ available

1, 2, 1, 1 ≤ 4, 5, 9, 6

↳ true

∴ T4 can be executed

available = available + allocation

= 4, 5, 9, 6 + 0, 1, 2, 1

= 4, 6, 11, 7

∴ The system is in safe state

safe sequence = T0 → T1 → T2 → T3 → T4

c) Request from T2 (1, 1, 1, 0)

request ≤ need

1, 1, 1, 0 ≤ 2, 2, 1, 2

↳ ~~yes~~ true

request ≤ available

1, 1, 1, 0 ≤ 1, 2, 2, 1

↳ true

available = (1, 2, 2, 1) − (1, 1, 1, 0) = 0, 1, 1, 1

allocation T2 = (1, 2, 3, 1) + (1, 1, 1, 0) = 3, 3, 4, 1

need T2 = (2, 2, 1, 2) − (1, 1, 1, 0) = 1, 1, 0, 2

T0 : need ≤ available
0,1,1,1 ≤ 0,1,1,1
↳ true
∴ T0 can be executed

∴ available = 0,1,1,1
+ 0,1,1,2
= 0,2,2,3

T2 : need ≤ available
1,1,0,2 ≤ 0,2,2,3
↳ false

T3 : need ≤ available
1,2,1,1 ≤ 0,2,2,3
↳ false

T4 : need ≤ available
1,2,1,1 ≤ 0,2,2,3
↳ false

T5 : need ≤ available
1,2,2,2 ≤ 0,2,2,3
↳ false

d) analysis of $T_1$'s request for (0,1,1,1)
request = 0,1,1,1
available = 0,1,1,1
request ≤ available
0,1,1,1 ≤ 0,1,1,1
↳ true

new available = (0,1,1,1) − (0,1,1,1)
= 0,0,0,0

need T0 = 0,1,1,1
need 0 > available

no thread at all can proceed.

Q10] page size = 1KB = $2^{10}$ bytes

a) address reference = 21205

page no. = $\left\lfloor \frac{21205}{1024} \right\rfloor = 20$

offset = 21205 %. 1024 = 725

b) address reference = 164250

page no = $\left\lfloor \frac{164250}{1024} \right\rfloor = 160$

offset = 164250 %. 1024 = 410

c) address reference = 121357

page no = $\left\lfloor \frac{121357}{1024} \right\rfloor = 118$

offset = 121357 %. 1024 = 525

d) address reference = 16479315

page no = $\left\lfloor \frac{16479315}{1024} \right\rfloor = 16093$

offset = 16479315 %. 1024 = 83

e) address reference = 27253187

page no. = $\left\lfloor \frac{27253187}{1024} \right\rfloor = 26614$

offset = 27253187 %. 1024 = 451

Q11] Logical address space = 2048 pages

page size = 4KB = $2^{12}$ bytes

physical memory = 512 frames

∴ page offset = 12 bits

a) Bits required in the logical address

no. of pages = 2048 = $2^{11}$ pages

∴ page no. field = 11 bits

logical address size = page no. bits + page offset bits

= 11 + 12 = ~~23~~ 23 bits

b) no. of bits required in physical address

frame no. bits = 9 bits

page size = frame size

∴ offset = 12 bits

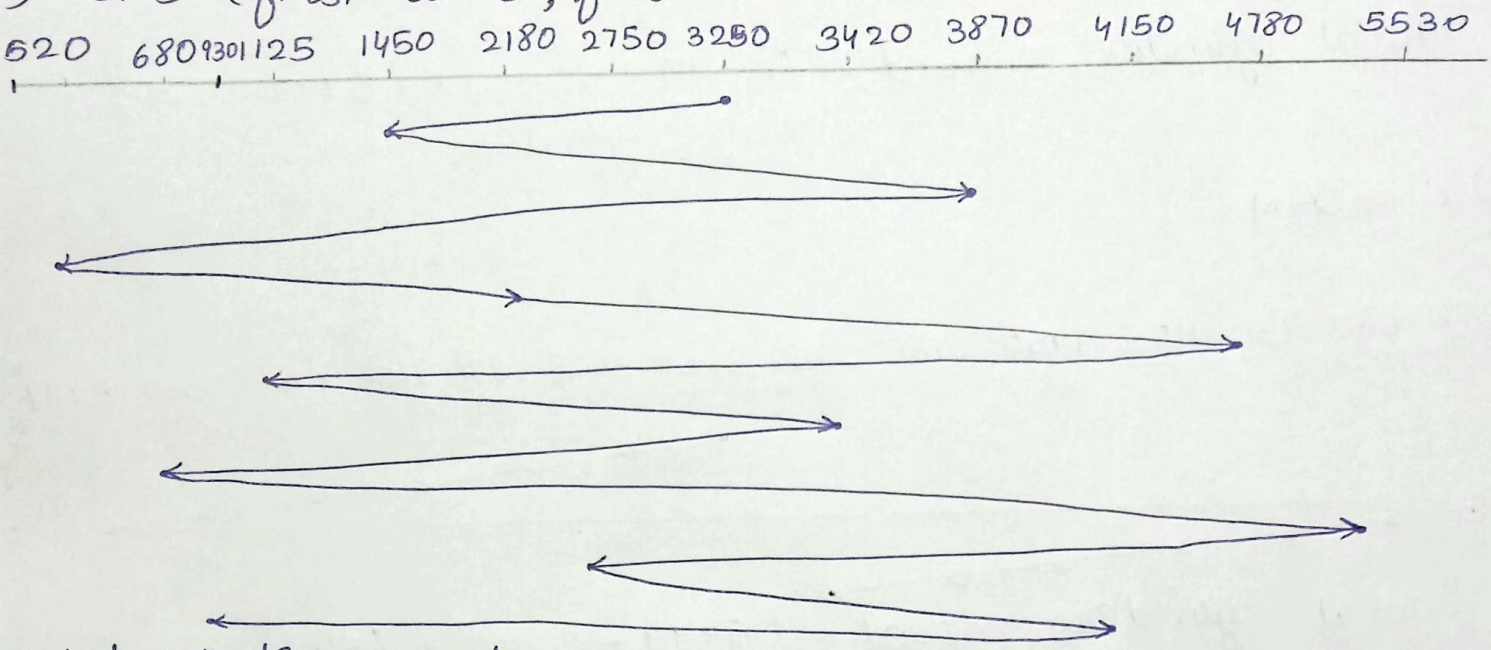
physical address = frame no. + page offset

= 9 + 12

= 21 bits

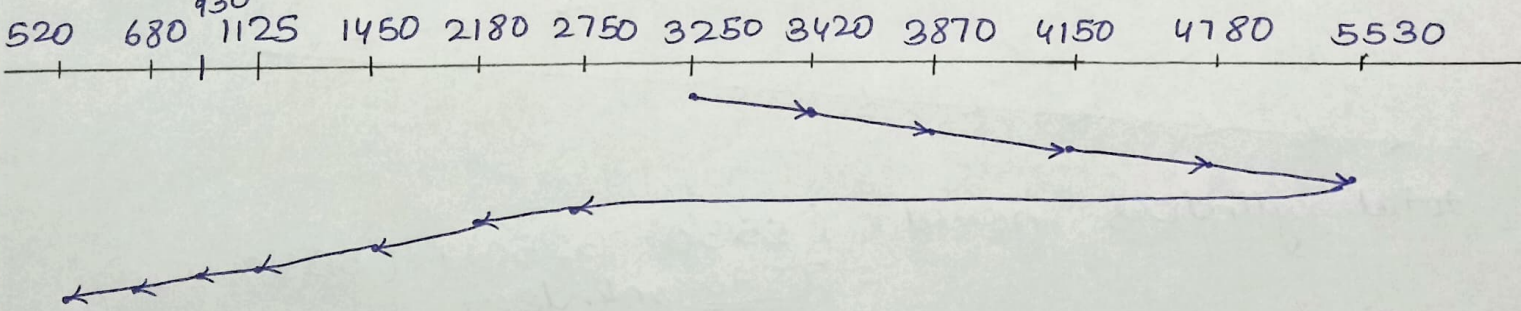
Q12] current head position = 3250
previous head position = 2940
cylinders = 0 to 5999
pending requests:
1450, 3870, 520, 2180, 4780, 1125, 3420, 680, 5530, 2750, 4150, 930

1) FCFS (first come, first serve)



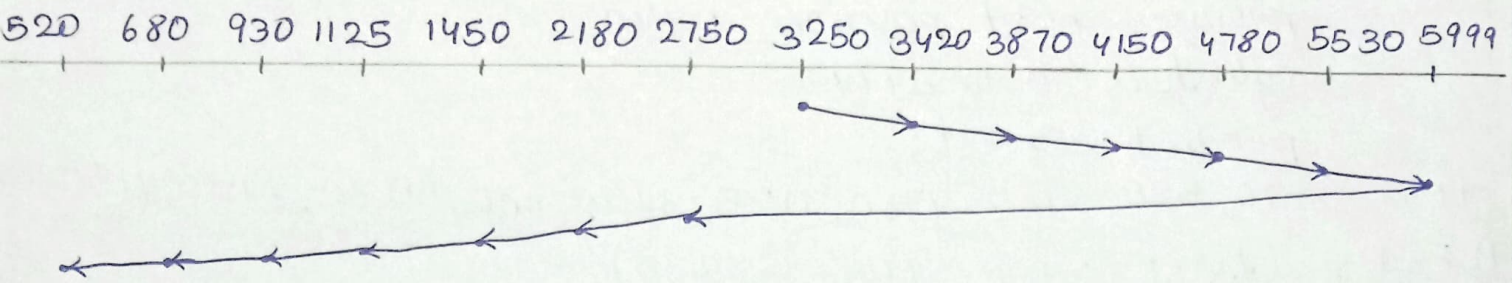

total cylinders moved = (3250 − 1450) + (3870 − 1450) + (3870 − 520) + (2180 − 520) + (4780 − 2180) + (4780 − 1125) + (~~3420~~ 4780 − 1125) + (3420 − 680) + (4150 − 930) + (5530 − 680) + (5530 − 2750) + (4150 − 2750)
= 32730 cylinders

2) SSTF



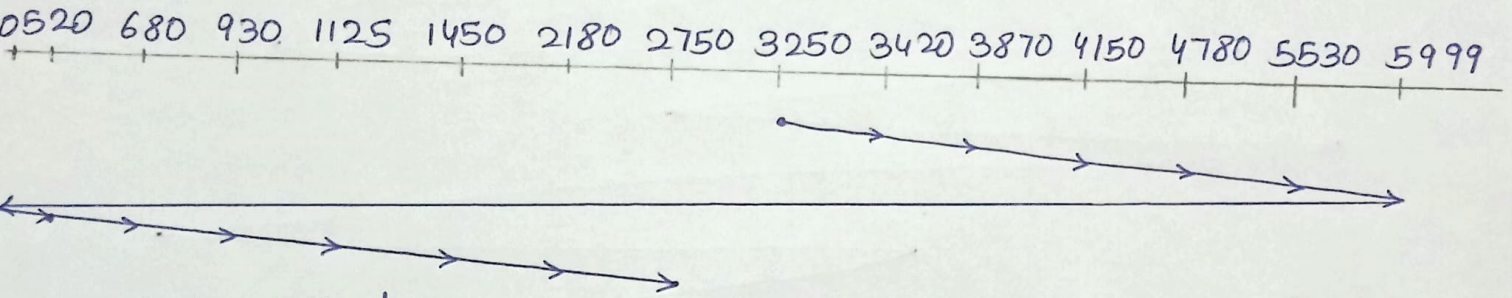

total cylinders moved = (5530 − 3250) + (5530 − 520)
= 7290 cylinders.

3) SCAN



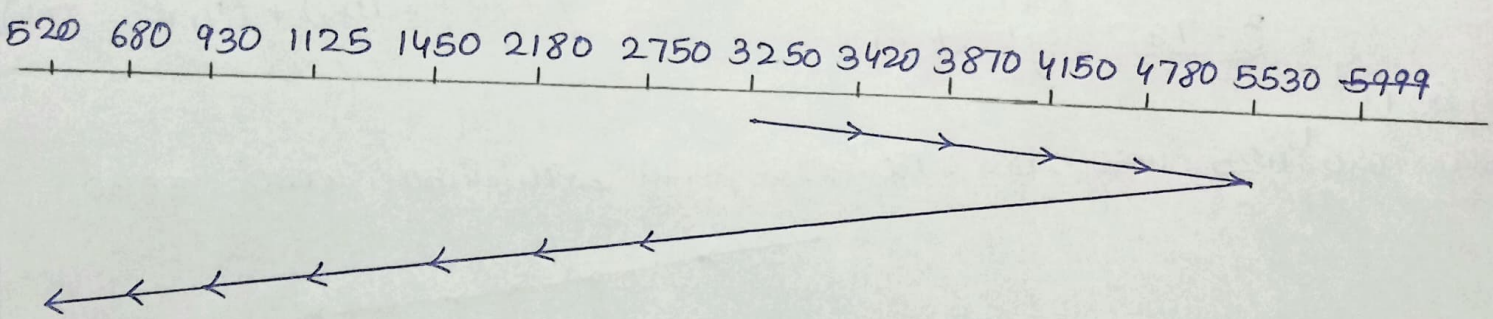

total cylinders moved = (5999 - 3250) + (5999 - 520)
= 8228 cylinders

4) C-SCAN



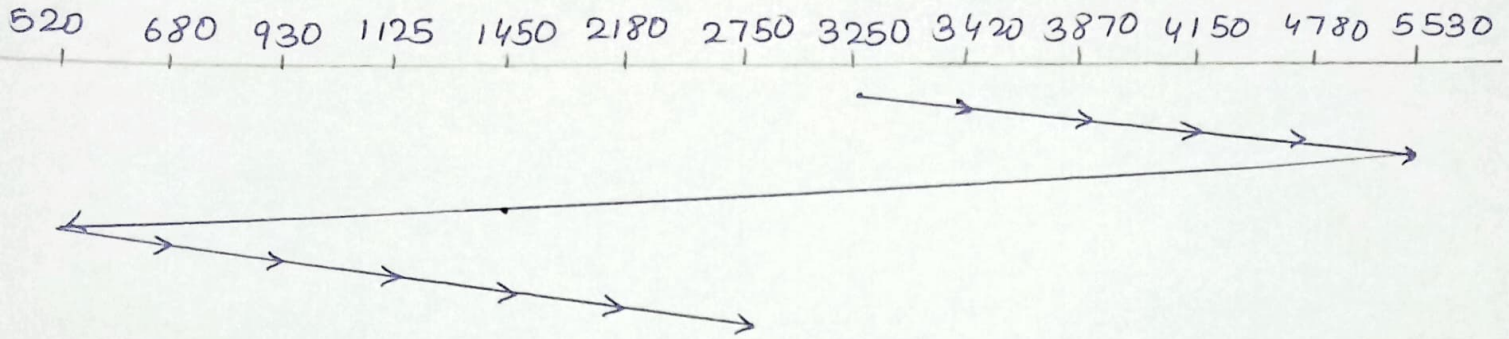

total cylinders moved = (5999 - 3250) + (5999 - 0) +
(2750 - 0)
= 11498 cylinders

5) Look




total cylinders moved = (5530 - 3250) + (5530 - 520)
= 7290 cylinders

6) C-Look.




total cylinders moved = (5330 - 3250) + (5530 - 520) + (2750 - 520)

= 9520 cylinders

| Algorithm | total head movement |
|---|---|
| FCFS | 32730 cylinders |
| SSTF | 7290 cylinders |
| SCAN | 8228 cylinders |
| C-SCAN | 11498 cylinders |
| LOOK | 7290 cylinders |
| C-LOOK | 9520 cylinders |

∴ Best performing = SSTF & LOOK
because of lowest head movement (7290)

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